V3 moneyness performance

Same 6 models · 3 companies · 4 regimes · 1.5-year monthly study · classified by call moneyness m = S/K

What this report is

- Companion to V3_1p5y_monthly_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets (n = 0) show em dashes and have no BEST.

Unchanged experimental design

- Same six models, three companies (SPY, AAPL, MSFT), four V3 regimes, 1.5-year lookback, monthly recalibration, V3 filters, Euler-corrected Heston step, LSM n_paths = 2000 / seed 42.
- Same option contracts as the main study (Monday nearest-ATM listed calls, DTE 7–60, $|S/K - 1| \le 10\%$). No model-specific re-sampling.
- Pooled n = 601 contracts per model (identical keys across models).

Moneyness rule (calls, m = S / K)

- DOTM: $m < 0.98$. OTM: $0.98 \le m < 0.995$. ATM: $0.995 \le m \le 1.005$. ITM: $1.005 < m \le 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \le 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008–2009.

Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{model} - C_{mkt})/C_{mkt})^2)}$. Ranking uses this only.
- MAE and Bias $ in option-premium dollars. $\text{Bias\%} = 100 \times \text{mean}((C_{model} - C_{mkt})/C_{mkt})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model × Moneyness RMSE% heatmap.
- Analysis 2 — Regime × moneyness: keep the four regimes separate, pool the three companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each regime.

Page map: p.1 method · p.2 sample counts · p.3–4 Analysis 1 tables · p.5 overall heatmap · p.6–7 Analysis 2 · p.8 findings.

Page 2 · Shared sample by moneyness · identical for all six models

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2008-2009 Crisis	8	30	72	34	6	150
2013-2014 Normal	0	5	142	3	0	150
2018-2019 Late-cycle	0	4	140	7	0	151
2019-2020 COVID	0	6	139	5	0	150
All regimes	8	45	493	49	6	601

Moneyness rule · American calls, m = S/K

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	8	1.3%
OTM	$0.98 \leq m < 0.995$	45	7.5%
ATM	$0.995 \leq m \leq 1.005$	493	82.0%
ITM	$1.005 < m \leq 1.02$	49	8.2%
DITM	$m > 1.02$	6	1.0%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1–M3

Pooled SPY + AAPL + MSFT and all four regimes. Bold green = lowest RMSE%.

Table M1 · DOTM · m < 0.98 · n = 8 · BEST = Heston

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	8	40.40	0.9541	+0.1029	+17.93	22.3%	
GARCH	8	39.58	1.0074	+1.0074	+31.78	19.4%	
Heston	**8**	**28.18**	**0.8867**	**+0.8250**	**+22.79**	**24.0%**	**BEST**
Merton	8	41.10	1.0019	+0.1650	+19.26	19.1%	
GARCH–Merton	8	51.18	1.5433	+1.5433	+44.92	22.2%	
Heston–Merton	8	28.99	0.9355	+0.9355	+23.63	20.6%	

Table M2 · OTM · 0.98 ≤ m < 0.995 · n = 45 · BEST = Heston

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	45	67.36	1.7884	+0.9067	+36.61	23.8%	
GARCH	45	72.16	1.8679	+1.5644	+48.95	28.8%	
Heston	**45**	**37.80**	**1.1721**	**+0.7247**	**+23.43**	**25.0%**	**BEST**
Merton	45	64.91	1.7235	+0.8110	+34.40	20.3%	
GARCH–Merton	45	86.75	2.3155	+2.0820	+66.51	27.6%	
Heston–Merton	45	38.86	1.1694	+0.6418	+22.94	19.8%	

Table M3 · ATM · 0.995 ≤ m ≤ 1.005 · n = 493 · BEST = Heston

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	493	51.70	2.0852	+1.3903	+34.04	27.1%	
GARCH	493	60.35	2.2439	+1.9522	+41.43	30.0%	
Heston	**493**	**48.25**	**2.0136**	**+1.6772**	**+35.44**	**28.5%**	**BEST**
Merton	493	48.60	1.9618	+1.2343	+31.00	23.4%	
GARCH–Merton	493	70.31	2.7868	+2.5402	+53.33	27.6%	
Heston–Merton	493	50.83	1.9345	+1.5683	+34.48	22.2%	

Bias $ = mean(model − market). Bias% = 100 × mean((model − market)/market).

Page 4 · Analysis 1 · Overall moneyness · Tables M4–M5

Pooled SPY + AAPL + MSFT and all four regimes. Bold green = lowest RMSE%.

Table M4 · ITM · 1.005 < m ≤ 1.02 · n = 49 · BEST = Heston

Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	49	41.26	1.4788	+0.7388	+18.69	28.6%	
GARCH	49	60.12	1.6708	+1.5690	+47.39	34.5%	
Heston	**49**	**30.37**	**1.2322**	**+0.8185**	**+16.27**	**31.2%**	**BEST**
Merton	49	38.91	1.3524	+0.6219	+16.37	24.9%	
GARCH–Merton	49	69.76	2.0516	+1.9321	+57.33	30.9%	
Heston–Merton	49	30.97	1.1431	+0.7327	+15.98	24.7%	

Table M5 · DITM · m > 1.02 · n = 6 · BEST = Heston

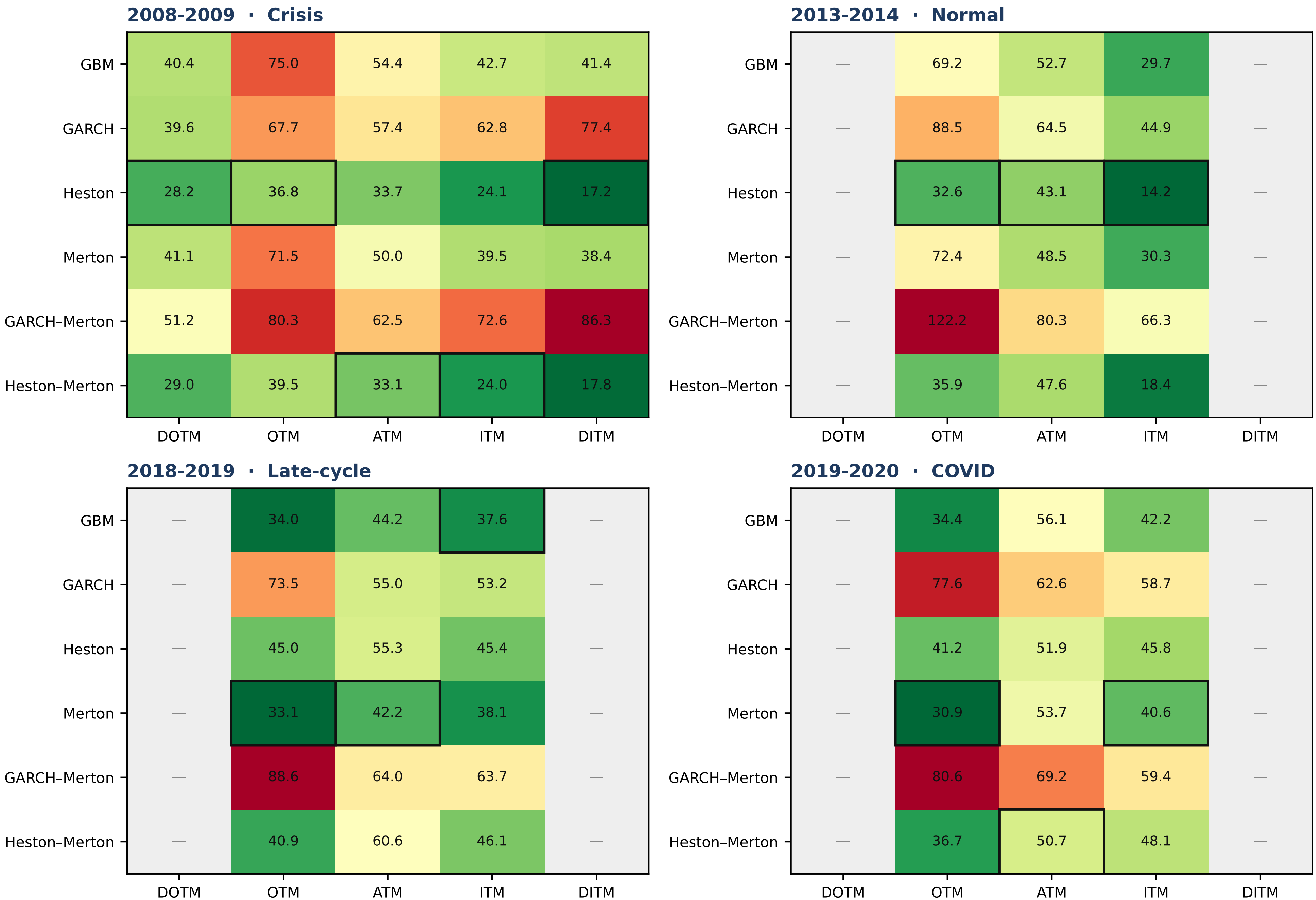
Model	n	RMSE%	MAE	Bias $	Bias%	Early-ex.	Mark
GBM	6	41.39	0.7738	-0.1103	+19.62	27.5%	
GARCH	6	77.41	2.4455	+2.4455	+64.89	28.8%	
Heston	**6**	**17.20**	**0.3569**	**+0.2325**	**+9.70**	**31.5%**	**BEST**
Merton	6	38.41	0.7773	-0.1722	+16.39	23.9%	
GARCH–Merton	6	86.27	2.9468	+2.9468	+73.23	21.2%	
Heston–Merton	6	17.83	0.3554	+0.3296	+11.51	22.5%	

ITM/DITM: S > K. DOTM/OTM: S < K. ATM: |S/K − 1| ≤ 0.5%.

Page 5 · Figure M1 · Model × Moneyness RMSE% · overall (pooled)

Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

Page 6 · Figure M2 · RMSE% by model × moneyness · each regime (companies pooled)



Outlined cell = lowest RMSE% in that regime × moneyness column. Grey = n = 0.

Page 7 · Analysis 2 · Does the ranking change with moneyness inside each regime?

Table M6 · Best model (lowest RMSE%) by regime × moneyness

Regime	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	Heston	Heston	Heston–Merton	Heston–Merton	Heston
2013-2014 Normal	—	Heston	Heston	Heston	—
2018-2019 Late-cycle	—	Merton	Merton	GBM	—
2019-2020 COVID	—	Merton	Heston–Merton	Merton	—

Table M7 · n contracts by regime × moneyness (shared across models)

Regime	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	8	30	72	34	6
2013-2014 Normal	0	5	142	3	0
2018-2019 Late-cycle	0	4	140	7	0
2019-2020 COVID	0	6	139	5	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that regime. Columns with n = 0 cannot change the ranking. Post-2009, ATM holds almost all of the Monday sample.

Does performance change with moneyness?

- DOTM (n = 8): BEST = Heston with RMSE% = 28.18, Bias% = +22.79.
- OTM (n = 45): BEST = Heston with RMSE% = 37.80, Bias% = +23.43.
- ATM (n = 493): BEST = Heston with RMSE% = 48.25, Bias% = +35.44.
- ITM (n = 49): BEST = Heston with RMSE% = 30.37, Bias% = +16.27.
- DITM (n = 6): BEST = Heston with RMSE% = 17.20, Bias% = +9.70.
- Overall, the moneyness ranking is stable: Heston is BEST in every non-empty bucket.
- Inside regimes, the BEST model also changes with moneyness in: 2008-2009, 2018-2019, 2019-2020.
- These comparisons use the Monday nearest-ATM sample, so ATM is the only well-populated bucket after 2009. DOTM/DITM results rest on few 2008–2009 quotes and should be read as descriptive, not as a high-powered test.